A spectral performance model for optical fibre networks

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This paper reviews the design and implementation of a software package that can be used to simulate the spectral performance of optical fibre networks. The model provides a novel, comprehensive solution for deriving the spectral dependence of both forward and return losses, through a diversity of routes in the simulated network. This enables a wide range of network designs to be modelled, and the network scheme can be overlaid directly on to actual geographical maps, thus providing a number of potentially valuable network design, planning, and maintenance tools.

1. Introduction

The design of optical fibre networks presents a new set of challenges to telecommunications network external planners (particularly access planners) who are more familiar with copper networks. Although the practical experience gained from implementing optical network field trials provides valuable guidelines in the design and planning stages [1, 2], there is a requirement for simulation models and planning tools which can structure the overall design strategy, and provide operational validation of a planner's proposed network scheme, within a rapid prototyping environment. Such tools might comprise a PC-based software system which addresses some of the potential needs of the optical network planner. These requirements could include the following:

- a technique for creating and modifying optical networks within design guidelines but without topological constraints,
- the determination of a number of key technical parameters of the optical network including the transmission loss power budget, return loss and general spectral behaviour of the network,
- a strategy for integrating geographical location maps into the planning process,
- a mechanism to allow the spectral performance parameters of each optical component to be specified using either theoretical values or measured data,
- a mechanism for optimising the layout of the outside plant with respect to cable routeing, available ducting and street furniture,
- a simple software user interface.

This list is by no means exhaustive, but it addresses most of the major factors which will determine the operational viability of an optical transmission network at the physical layer. The important issues of network whole-life cost, reliability at the physical and information transport layers, and resilience to intrusive damage, must additionally be addressed once it has been established that a particular topology is viable in terms of transmission capability. This is discussed elsewhere in this issue, and the literature [3—6] indicates that there are a large number of access network performance modelling tools under development. None of these seems to provide the functionality described in the requirements list above. For instance the proof-of-concept tool developed by Bergstrom et al [3] neglects to provide a means of determining the return loss of optical networks, and it does not take into account the diversity of power budgets which needs to be considered due to either multiple path or multiple wavelength operation. Ybema's software system [4] provides a means of optimising the routeing of cables in a green fields situation, but does not take into account the actual geographical location map. The NETCAP system developed by Jack et al [5] addresses the problem of how to expand the capacity of a network to meet future service requirements based on the minimisation of the cost of location of outside plant components. In short, none of these software packages tackles the problems of integrating the geographical layout of the outside plant with the real performance data of the optical components, and none provides a means of analysing the forward and return losses and power budgets, based on specific paths through the optical networks.

Although the simulation software has been used principally to develop and test the transmission loss algorithms, care has been taken to provide a user-friendly interface which could provide the basis of a transmission planning tool. The user can visually create or modify optical networks simply by picking up optical elements from a menu, and placing these elements at the required positions on a geographical location map. This is produced by scanning the original paper-based street or ducting maps and provides a background on which the user can layout an external optical network. Automation of the process to find cost-optimal duct routes between the nodes of a network could then be accommodated using for example the algorithms described in Cusack and Cordingley [7] and Paul and Tindle [8]. In the current implementation the planner is required manually to select the location of ducting routes and street enclosures for the outside plant.

The optical elements provided to the user can also be represented within the package as objects which are either derived from a primitive generic ancestor or from a previously defined optical element. Hence by using an object oriented (OO) design methodology [9], the functionality of the package could be extended either in response to changing requirements or as the need for modifications arise. Although the initial parameters of interest in the network would be the optical power budget and return loss, the OO methodology also permits the spectral characteristics of each device to be included as attributes in the object description.

2. System description

A block schematic of the hardware and major software modules within the package is shown in Fig 1. The hardware consists of a flatbed scanner connected to a 486 PC. The scanner enables both map images and other data (e.g. optical device spectral characteristics) to be digitised and held in files on the PC. Excluding the proprietary scanner software, the functions of the package are divided into two main software modules — the ‘analysis’ and ‘capture’ modules which are also illustrated in Fig 1. These appear to the user as separate tools within the package and they are indeed independent in operation. The system has been constructed using the object oriented PASCAL programming language within the Windows 3.1 operating system [10].

The analysis module is a tool for the creation and subsequent performance evaluation of optical networks. It appears to the user as a menu-driven window to which a range of commands from the user can be directed. These commands constitute one set of inputs to the module. The other important inputs to the analysis module consist of location maps obtained from the scanner in the form of Windows 3.1 format binary image files. Additionally the user can specify options from a set of drop-down menus to supply further data to the analysis module. In response to the various inputs, the analysis module provides a number of outputs. These include a visual display of both the location map and user-generated schematics of optical networks, and, most importantly, the spectral behaviour of the optical network forward and return losses.

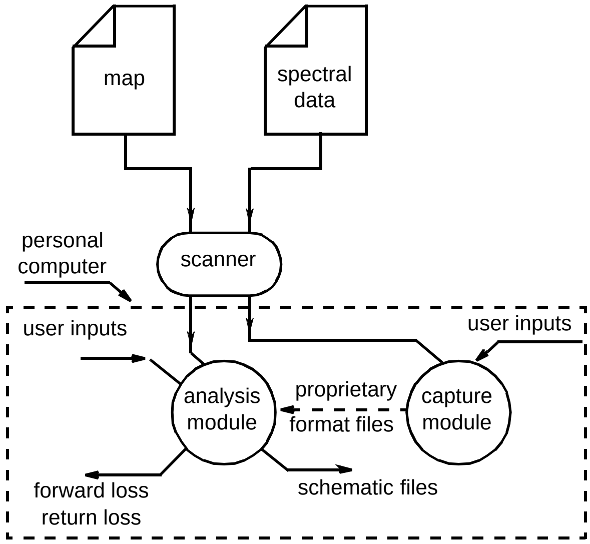


Fig 1 Block schematic of the system.

To simplify further the use of both system modules by designers or planners, a context-sensitive help system has been integrated into each one to assist two categories of user. For a user completely unfamiliar with the Windows 3.1 environment on which the package has been implemented, the help system links into the Windows 3.1 help engine [10] to provide demonstrations of how to operate within this environment. For users experienced in the use of Windows 3.1, but unfamiliar with how the package works, the help system provides a self-documenting guide to the operation of the package and the resource models used. At any point in the design or planning process the help system can be accessed by the user via a menu option.

To construct a network on screen, the user selects the menu option ‘resource’. This choice activates the optical resource editor sub-module which causes it to create an instance of the selected resource and then place it at the position specified on the screen. In turn, the ‘network schematic’ sub-module is invoked (which is transparent to the user) whenever links are placed to create interconnections between nodes. This ensures that an incremental description of a network is built up as optical resources are placed by the user.

A typical optical access network planning cycle using this package would start with the selection of a target location on which to place the optical resources (e.g. nodes and links). Location maps can be obtained from a digitised image library, converted data file, or scanned directly using a desktop scanner. The image in Fig 2 is an example of an

Fig 2 Location map entered from scanner.

image of a location obtained by directly scanning a paper copy of a map into the software package. Once the location map has been entered into the package, the map can be used by the network designer as a visual aid in the locating of optical resources exactly where they are required. Furthermore an overview image of the location, which is generated at the same time as the map, is then entered into the package and enables the designer easily to navigate the entire map. Once the optical resources have been placed on the location map the network designer has the option of modifying the default characteristics according to design requirements. Figure 2 further illustrates the appearance of a location image after a number of optical resources have been placed. At this stage of design the digitised characteristics obtained from the capture module can be associated with the various optical resources placed on the location map. During the resource placement process a number of editing functions are available to the system user that make it straightforward to effect design changes or to undo errors in the network schematic. Analysis of the network yields forward and return spectral losses together with the ranging delays of each loss component. Because of the definition of network structure the analysis of several independent networks at the same location can be carried out. Continuity between network design sessions is of course assured by the ability to save, retrieve and update network implementation schemes.

3. Details of the analysis module

The key task for the analysis module is to compute the forward and return losses of the planned optical network scheme. In general terms, the simulated optical network consists of optical node resources (joints, splitters, transceivers) interconnected by fibre links. Within such networks any interconnection of optical node resources by optical fibre links is allowed subject to two conditions:

- the terminals of each node can be connected to at most one other node,
- a terminal of a node must not be connected back to another terminal of the same node.

In order to analyse networks subject to these criteria a precise and unambiguous specification of such optical networks is required. Backus-Normal Form (BNF) was therefore chosen as a formal method of representing the structure of optical networks within this package [11]. BNF is a meta-language intended for the concise and unambiguous representation of hierarchical relationships. It facilitates the precise description of the physical structure of an optical network by clearly indicating the nature of the interconnections that define the topology of the network.

Each node in a network can be represented by an optical resource which is either a primitive or constructed from other primitives and whose parameter of interest is the power transfer characteristic. A network can therefore be defined by three production rules stated in BNF :

network ::= <node> (<link> <sub network>)*
subnetwork ::= <node> | <network>
node ::= transmitter | receiver | cable_tray | splitter |
combiner | tap | attenuator | amplifier

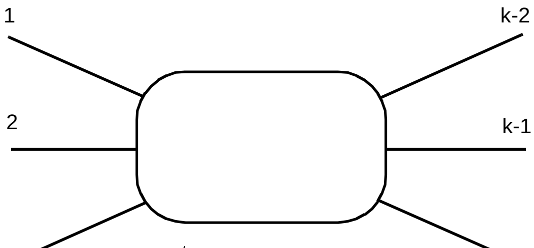
The left hand side of a BNF equation (or production rule) introduces a new BNF variable called a non-terminal which is an item whose definition is pending. The right hand side of a BNF production describes the structure of the left hand side in terms of other non-terminal symbols, terminal symbols (which represent primitive elements) and certain operators. The basic BNF operators relevant to the definition of an optical network are:

- the symbol | is an alternation (one or the other of the listed symbols),
- the symbol * is an iteration (zero or more of the listed symbols),
- the symbol ( ) represents operator priority.

Hence the BNF production rules for the optical subscriber loop provide a recursive definition of the possible structures of an optical access network and the production rules stated above can be used to describe tree, star, bus, ring and mesh topologies. From these production rules it may be observed that a network can consist of a node connected by one or more links to a number of subnetworks. Each subnetwork can in turn comprise either a node or it can have the structure of a network. At the level of the node then the possible choices of primitives that can be utilised are implemented by the third BNF rule. This list of choices represents the primitive optical elements that can be handled by the package at the present time. These elements represent distinct instances of particular classes of optical resources. Thus, although a network may (and usually does) have more than one terminal element of a specified kind, each occurrence of the terminal element is a distinct occurence of the class to which it belongs.

Having specified the overall structure of a network using the BNF notation, the form and behaviour of the individual nodes and links of the network (the optical resources) will then be specified using the principles of object oriented technology. Each node was derived from a generic primitive object which encapsulated the essential characteristics (as object fields and methods) of an optical resource. These essential characteristics refer to behaviour that is common to every optical resource. Following a hierarchy of increasing specialisation every other optical resource was constructed from, and inherited the properties of, this generic ancestor. Each level in the specialisation hierarchy led to a new object class. Such an object class served as a template for particular instances of that kind of object. Implementing network resources as objects makes it possible to rapidly update the basic set of resources available to a user. For instance, having defined a three-port coupler from the generic resource, a wavelength division multiplexing device [12] can be described in terms of the coupler by extending its properties to include wavelength selective behaviour.

The key characteristic used in the computation of the forward and return power losses is a power transfer matrix $T(\lambda)$ [13] which specifies the wavelength-dependent insertion loss $T_{ij}(\lambda)$ between any pair of node terminals i and j. In this case the upstream terminal is denoted by i and the downstream one denoted by j, where the special case of $i = j$ denotes a reflection loss at terminal i(or j). Therefore, as illustrated in Fig 3, if there are k terminals connected to a particular node, then a power-transfer matrix with k^2 terms can be formed at each wavelength of interest. Within the package the user can easily specify or change each of the terms in the power-transfer matrix of any optical resource. The BNF specification of a network together with the power transfer matrix of each of its resources are used in all the power-flow computations.



$$T = \begin{pmatrix} T_{11} & T_{12} & \dots & T_{1k} \\ T_{21} & T_{22} & \dots & T_{2k} \\ \vdots & & \ddots & \\ T_{k1} & T_{k2} & \dots & T_{kk} \end{pmatrix}$$

Fig 3 Optical resource and power transfer matrix.

In order to compute the forward losses, the BNF specification is first used to trace all the possible paths between every transmitter and each terminal element in the network. Each time a resource is encountered on a path, the term in the power-transfer matrix of that resource indexed by the upstream and downstream terminals is used to compute the node insertion loss. Designating the path length of the link between any two successive nodes on a traversed path as L_{path}, the link attenuation per unit length as $\alpha_{path}(\lambda)$ and the insertion loss between terminals i and j as T_{ij} node, then the forward loss (FL) will be given by the expression:

$$FL(\lambda) = \sum_{paths} \left(T_{ij}^{node}(\lambda) + \alpha_{path}(\lambda)\, L_{path}\right) \qquad \text{...... (1)}$$

The traversal algorithm takes the BNF description of an optical network and returns an enumeration of each path in the network. Thus, between any pair of optical resources in the network, the wavelength-dependent power loss over the paths found by the traversal algorithm is given by equation (1). The number of paths which will be found is equal to the product of the number of transmitters of optical power and the number of terminal nodes.

Modelling of the return loss is undertaken in a similar way to the forward power flow. Each point source (optical resource) of reflection within the network is modelled as a pseudo-transmitter. Each pseudo-transmitter is then assumed to launch power into the network at a level equal to the return loss at the location of the pseudo-transmitter point. In general, if R_k^n denotes the return power level at node n due to power launched from a transmitter k, then the return power level at any terminal node can be obtained by applying equation (1) to return paths defined in the direction of reflected power resulting from network traversals with node n as the starting point. Therefore, if a terminal node can be reached directly on the reflection path of a node n, the following expression gives the return loss (RL) at the network terminals with t transmitters:

$$RL(\lambda) = \sum_{k=1}^{t} \left(R_k^n(\lambda) + \sum_{return\_paths} (T_{ij}^{node}(\lambda) + \alpha_{return\_path}(\lambda) L_{return\_path}) \right) \quad \text{...... (2)}$$

Equation (2) enables the various components of the reflected optical power at each of the terminal nodes of an optical network to be calculated. These components result from reflections at node n due to launched optical power from each of the transmitters in the network. Equation (2) can be interpreted in the same way as equation (1) with *return_ path* substituted for *path*.

The reflected power components in an optical network can form the basis of an OTDR simulation. Taken together, the set of events which occur due to reflections in a network can readily be used to create a reference OTDR trace. Later comparison of the reference trace with an actual trace can assist the process of fault location.

Reflection losses, also known as Fresnel losses, occur whenever there is a change of refractive index along any path in an optical network [14]. Such changes of refractive index typically occur at joints between fibres or between fibres and other network resources. Using carefully designed mechanical joints or fusion splices, it is possible to reduce the return losses to negligible levels (lower than −40 dB) whenever fibre joints are made. The design of the optoelectronic components used at the terminals of the optical network also greatly affect the magnitude of reflections back into the network. Cable breaks also give rise to reflections whose amplitude depends on the exact nature of the break [15] and whose accurate location (e.g. by optical time domain reflectometry — OTDR) can be a powerful aid to network fault diagnostics and repair. In the model, fibre breaks are simulated by a pair of pseudo-transmitters which launch optical power back into the network at both ends of the broken fibre. In order to determine the effect of the break, equation (2) may be used with R^b in place of R_k^n , yielding the following expression for return loss due to a cable break in a network:

$$RL(\lambda) = R^b + \sum_{return\_paths} (T_{ij}^{node}(\lambda) + \alpha_{return\_path}(\lambda) L_{return\_path}) \quad \text{...... (3)}$$

The return loss information at each network terminal is also presented as a pair of spectral characteristics representing the best case and worst case losses along the path of interest. As the optical power launched from a transmitter spreads throughout the network each node encountered becomes a pseudo-transmitter. The pseudo-transmitter then reflects power back to all the network terminal elements that can be reached directly on its return (reflection) paths. In order to obtain the results of any return loss computation, the package user can specify the nodes at which power is launched into the network, the node from which it is reflected, and the node at which the return power is to be measured.

4. Operation of the simulation package

After the completion of an optical network scheme as described in section 2, the network designer can undertake an exhaustive analysis of both the forward and return spectral power losses in the optical network. In addition, from the information provided by the package about the spectral behaviour of a network, then the forward and return power losses at any specific wavelength can be determined. The schematic in Fig 4 illustrates a 32-way passive optical network (PON) based on two sets of wavelength-flattened couplers, whose spectral performance data has been scanned or otherwise loaded into the model. In this network, the two couplers respectively provide a 4-way and an 8-way split of the incoming optical power. This diagram is actually a screen dump taken from the software package, with the location map background image optionally suppressed in order to show the optical network schematic more clearly.

Before any power-flow computations are carried out the package undertakes a complete path listing of the network. This generation of a path listing is a task which is common to all the power-flow computations. Although the results obtained from this step are not visible to the planner, they are used in identifying analysis paths later in the power-flow calculations. As described in section 3, the path listing is generated by the traversal algorithm which takes the BNB description of the network and returns an enumeration of each path in the network. The path enumeration describes the routeing of optical power between each transmitter and every other terminal element. By separating the path enumeration task (which is common to all power budget analysis steps) from the analysis of the power budgets, the user can concentrate the analysis on specific parts of an optical network. This limits the amount of data generated

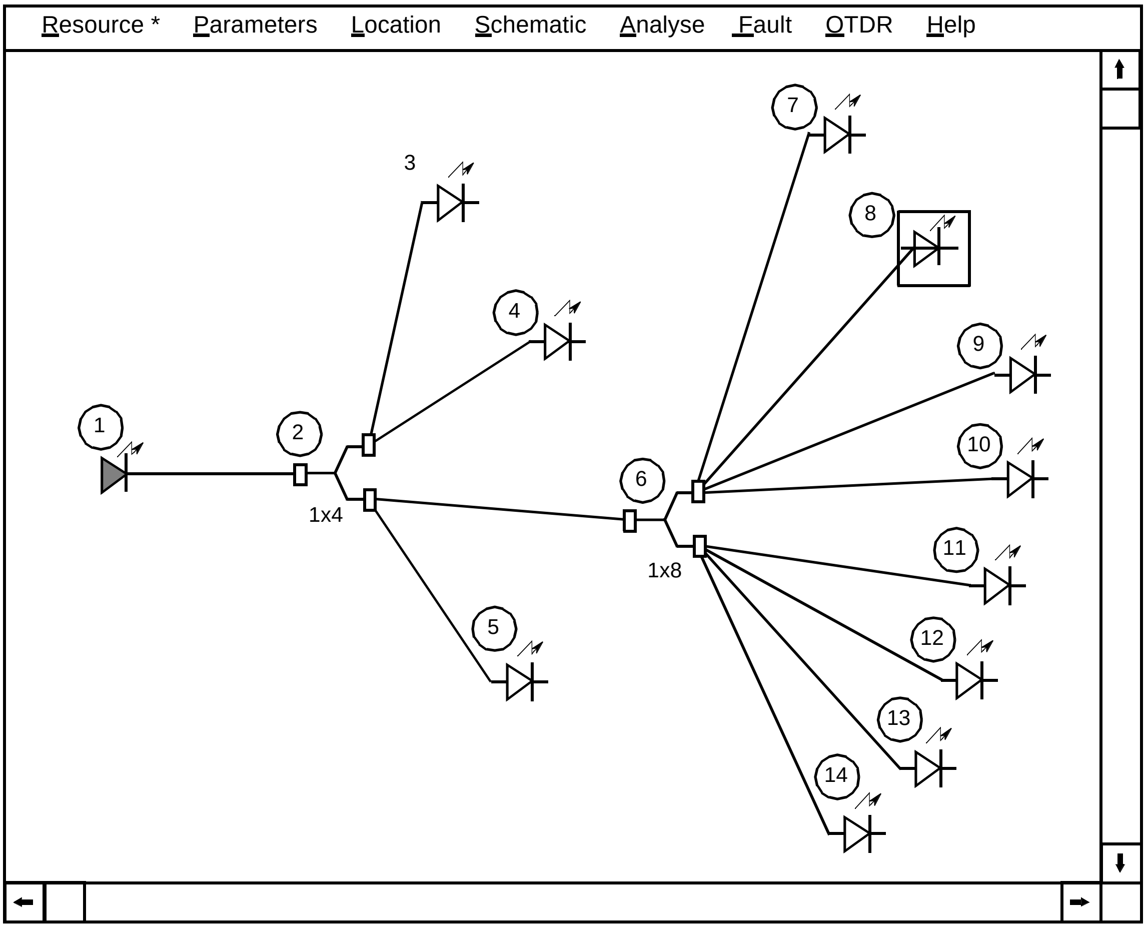


Fig 4 Screen showing an optical access network incorporating wavelength flattened couplers.

during an analysis phase. In order to specify a particular path for analysis, the user is required to change the state of the nodes on that path. Most nodes in a network can be set to any one of three states (or certain combinations of these) for the purpose of analysis:

- transmission,
- reception,
- reflection.

While only transmitter nodes can be set to the transmitting state, other nodes can be designated either to be receivers or reflectors, or both. Nodes in the transmit state appear within a solid box in the display (see Fig 4), nodes in the receive state appear within a bounding rectangle and finally nodes in the reflect state appear as flashing icons. In Fig 4, node 1 is in the transmitting state while node 8 is in the receiving state. In this case the only path in the optical network for which a full forward loss computation will be carried out is the one which originates at node 1 and terminates at node 8. The output screen dump following the forward spectral analysis on this path is shown in Fig 5. This screen illustrates the network report that can be generated by user request upon completion of a spectral analysis. The report enables each of the analysis paths in the network to be examined with respect to power flow, and, as seen in Fig 5, the information about the path spectral analysis is presented as both tabular values and as spectral characteristics displaying 'best' and 'worst' case curves. The user can alter the simulated wavelength range, as well as the assumed source launch power and receiver sensitivity, from the initial defaults provided by the path report. Any changes made to these default values by the user will be recorded in both the tabular values and the spectral display. The tabular values in the network report shown in Fig 5 also display the best and worst-case forward power losses as well as the total power margin for a link budget of 45 dB (i.e. 0 — 45 dB) on the selected path over a range of wavelengths. Negative margins in the table are indicated in red on the screen by the package and they represent those wavelengths at which the selected network path fails to meet the link budget requirement of 45 dB. This is an important feature of the package since it allows the planner/designer to identify which areas of an optical network fail to meet the necessary power budget requirements, at either a specific wavelength or over a desired wavelength range.

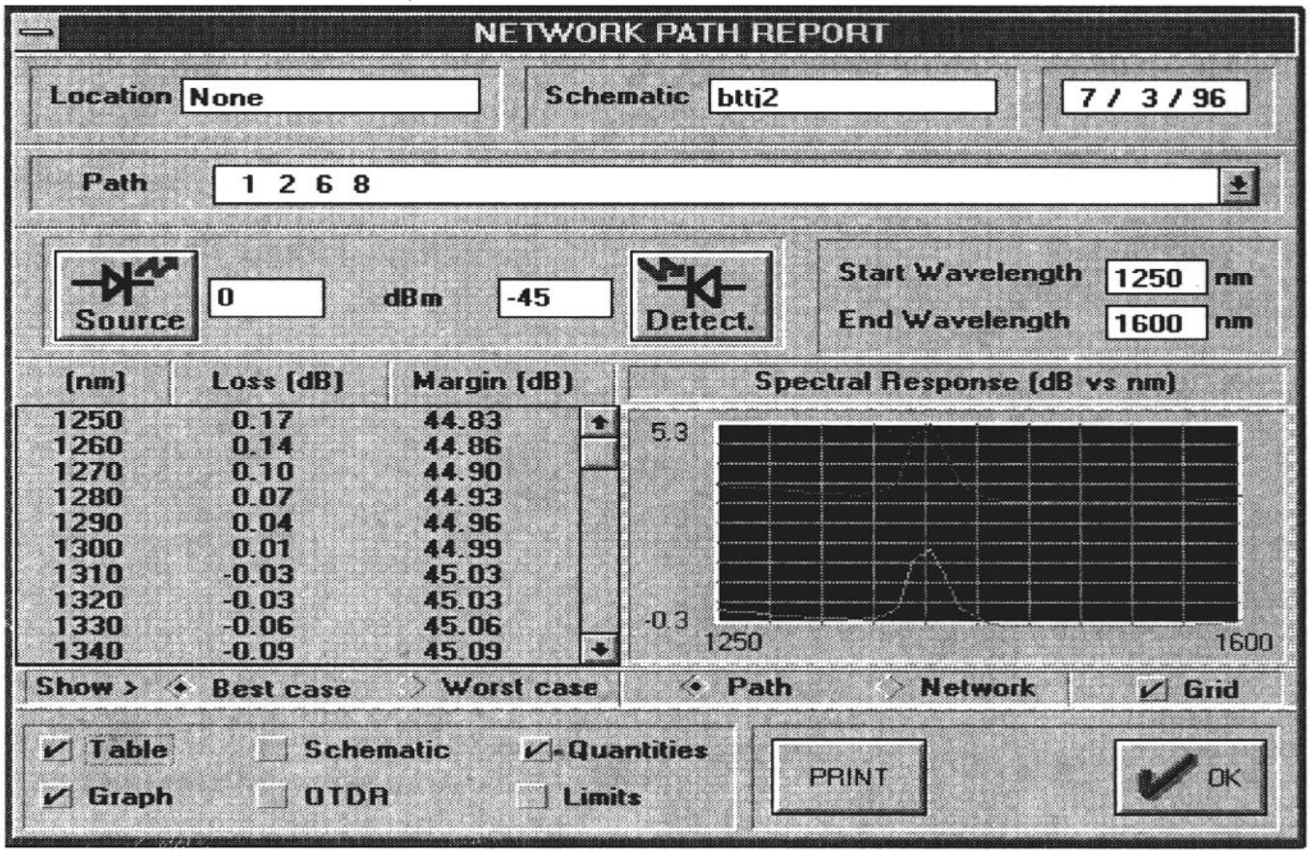


Fig 5 Screen showing the forward loss characteristics of the network of Fig 4.

Once the common task of path enumeration has been completed then the process of undertaking a return loss spectral analysis proves similar to that for a forward power budget analysis. A further example network is shown in Fig 6, which, for the purpose of clarity, is not a direct screen dump. This optical network has been constructed using a series of standard 2 × 2 couplers in order to demonstrate specifically the operation of the package in relation to optical return loss. Figure 7 shows two different sets of simulated return loss spectral characteristics for the network of Fig 6. Figure 7(a) displays the best and worst case return loss spectral characteristics along the network path <1, 2, 3, 5, 7>. To generate these characteristics, node 1 was designated to be simultaneously a transmitter and a receiver, while node 7 was designated as a reflector. The spectral characteristics shown in Fig 7(a) are obtained from the calculated ‘best’ and ‘worst’ case power levels arriving back at the transmitter (node 1) due to the reflection at node 7. The versatility of the package is demonstrated in Fig 7(b) which illustrates how it can be used to investigate the effect of return losses from nodes which are external to the path that links a transmitter to a terminal node. This set of spectral characteristics indicates the effect of the power reflected from node 7 on another node (node 6). To perform this calculation, the user indicates node 7 to be a reflector of optical power, node 6 to be a receiver and node 1 to be both a transmitter and a receiver of optical power.

5. Conclusions

The transmission simulation model which has been described could provide an important element in a software system for designing and planning optical fibre networks.

Additionally, the capability to calculate the effects of reflective components and fibre breaks allows the simulation of OTDR interrogation of the network, with potential applications in fault location and network maintenance. In its present form, the software package seeks to address several major requirements. Firstly, it provides a valuable tool for continuing research into optical subscriber network design. Secondly, it could assist in the development and evaluation of optical network. rule planning since it can be used by network planning rule designers and policy-makers to test the feasibility of a number of network configurations directly on maps of target locations. Thirdly, it meets some of the generic requirements of an optical fibre network tactical planning tool, potentially enabling external planners to rapidly construct and test their schemes directly on target location maps. Finally, the object oriented programming methodology has already been shown to be a robust and effective method for modelling optical networks [7] and in

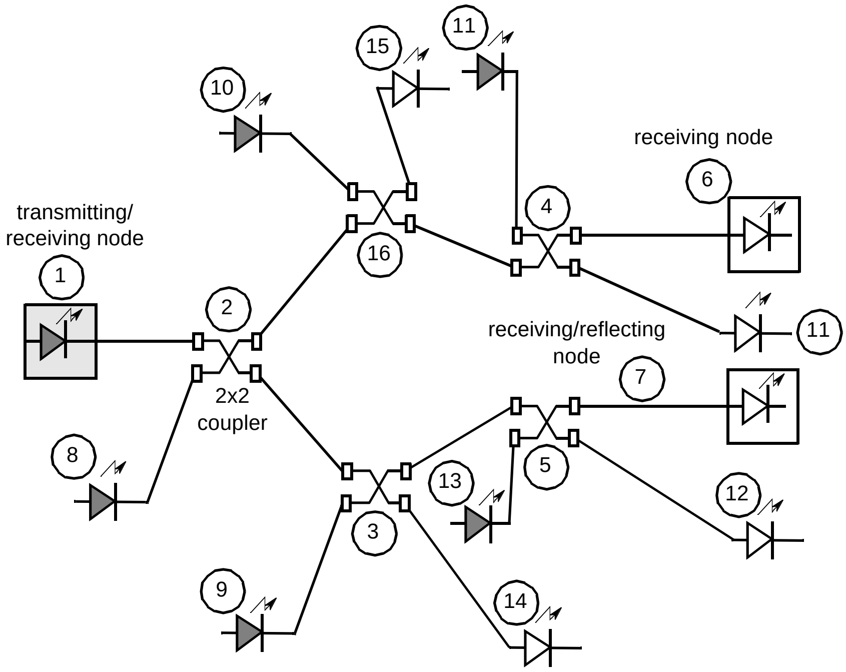
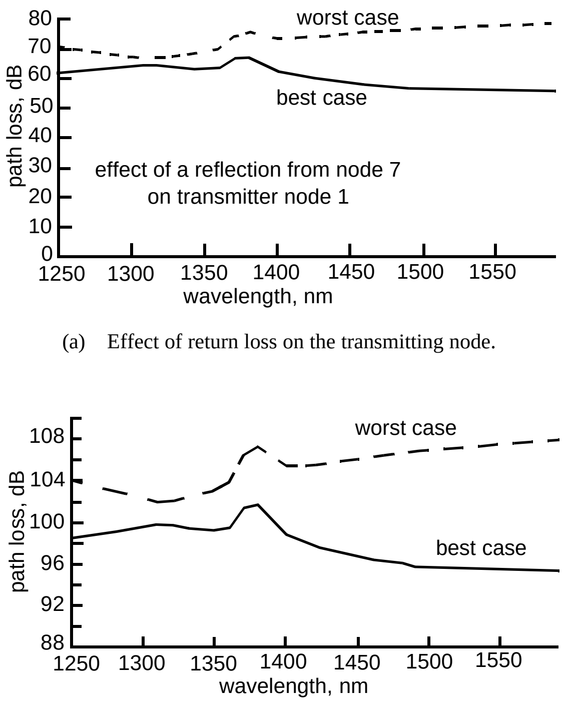


Fig 6 Passive optical network constructed from 2 × 2 couplers.



(a) Effect of return loss on the transmitting node.

(b) Effect of return loss on node 6.

Fig 7 Spectral return loss characteristics.

this case it will allow new types of optical resource easily to be included into the basic set, as the need arises.

The methodology has the advantage of supporting a modular software upgrade path as requirements change. Thus the functionality of the package could be extended by the creation of new class hierarchies or the modification of existing ones rather than requiring a fundamental reconstruction of the existing software system. Typically, such a strategy could extend the package to incorporate optical network diagnostic functions (for instance, comparison of measured and simulated OTDR response), duct-route optimisation, network construction bill of materials, and whole-life costing.

A network diagnostics capability could readily be integrated into the package; however, optimisation, build and life-cycle costing would need to be treated as longer term developments, and, as argued elsewhere in this issue, would be better treated as separate modules in a distributed network simulation system, sharing a core of performance and configuration information.

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17 Senior J M, Asumu D E and Finegan T: ‘A software package for designing/planning fibre networks in the subscriber loop’, Fibre and Integrated Optics J, 15, No 1 (1996).

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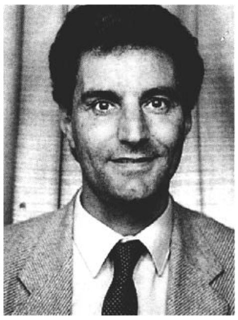

John Senior graduated with an honours degree in Physics from the University of Birmingham in 1973, and subsequently obtained a masters degree in Communication Engineering from the University of Manchester Institute of Science and Technology in 1975. He was then employed as a communications engineer with GEC Telecommunications and then Plessey Communications and Data Systems Ltd prior to joining the then Manchester Polytechnic in late 1979, where he established a research activity in optical fibre communications and networks which has grown to include significant collaborations with BT Laboratories. Since 1989 he has been Head of Department for the Department of Electrical and Electronic Engineering at the now re-named Manchester Metropolitan University, and he was subsequently awarded the title of Professor in 1990. In addition, he is now the Director of the Centre for Communication Networks Research within the Department, which undertakes a broad range of research activities concerned with communications systems and networks implementation and performance issues.

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John Mellis leads the Network Engineering Plant and Pilots unit at BT Laboratories, which specialises in the performance, planning, and pilot trial of network infrastructure components, cables and systems. He joined BT in 1986 to engineer network demonstrations of coherent optical transmission, and from 1989—91 was with BT&D (now HP Optoelectronics), commercialising optical amplifier and tunable laser products. From 1983—86 he was at STL, Harlow (BNR Europe), developing electro-optic/acoustic-optic signal processors. He received a doctorate from the University of St Andrews in 1983, for the mathematical modelling of CO_2 laser kinetics, and graduated first-class in Applied Physics (with Operational Research) from the University of Stratchlyde in 1978. He is chairman of the ETSI expert group on passive optical components, sits on the technical advisory committee of the European Conference on Networks and Optical Communications, and is a Member of the IEEE.